

Past Paper (2020)

Semester Sixth (6th)

~& (i) ~

What is closed Process ?

closed Process is the idea that the flow path of Product and materials could be operated as closed system, if a system can exchange only energy with the surroundings but not matter, is called closed system.

Example :-

Water placed in a closed metallic vessel. and when we boil water with a closed lid the heat can exchange but matter can not.

~& (i) ~

What is Partition Function :-

in Quantum mechanics, All the information about microscopic system such as position, velocity, momentum, energy etc. is stored in wave function ψ . In Statistical mechanics also there exist a function which contain all information about macroscopic system. it is that function is known as Partition function. it is dimensionless quantity. it is how the energy of system of molecule is partitioned among the molecules. The value of Partition function depends upon molar mass, temperature, molar volume, the intermolecular distance, molecular motion and inter nuclear force. it is denoted by symbol " Q ".

Q. Define 2nd Law of Thermodynamics?

This law may be expressed in variety of equivalent ways, this law states that, "it is impossible for a system to undergo a cyclic process which can convert all amount of heat into an equivalent amount of work, without producing other change in the system, or its surrounding."

Clausius statement :-

"We can not construct a heat engine or machine which can transfer heat energy from cold body to hot body or from low temp. to high Temp. without the expenditure of energy."

Application of 2nd law of thermodynamics :-

it helps in determine the direction of energy transfer.

it helps to predict whether a given process can occur spontaneously or not.

it helps to know the equilibrium position of the system.

it also helps to find the maximum fraction of heat that can be converted into work.

iv)

Define this eq.

What is meant by Degree of Freedom. The state of gas can be specified if we define state of each molecule of the gas. The state of each molecule can be specified by defining its position and velocity such that we must specify f -Positional and f -velocity coordinates, where f is the number of degree of freedom of molecule, Total No. of degree of freedom are,

$$f = 3 \times \text{No. of Atoms in molecule}$$

The minimum number of independent coordinates needed to specify the position and configuration of thermo-dynamical system in space is called Degree of freedom.

v)

Describe Most Probable Velocity ?
Solved in Paper 2021 " Question 5 "

vi)

What is Cage effect ?
Solved in Paper 2021 " Question 6 "

vii)

Reaction in Solution Phase are faster than gases phase why ?
Solved in Paper 2021 Question 07

(viii)

What is Law of Conservation of energy?

The law of conservation of energy states that energy can neither be created nor destroyed. But it can be changes from one form to another form it is called law of conservation of energy.

OR

The total energy of an isolated system are remain constant.

Consider a gas in cylinder having a piston. The internal energy of gas is E_1 . The heat " Q " is given to the system. Now internal energy of system becomes E_2 . The change in internal energy is given as.

$$\Delta E = E_2 - E_1$$

We have to do work on piston to keep it in its original position, thus energy change is equal sum of heat and work.

$$\Delta E = q + w$$

$$\Delta E = q + P\Delta V$$

It is mathematical form of first law of thermodynamics.

~~adix~~ Ben

What is difference B/w Spontaneous and NON Spontaneous Process ?

Spontaneous Process :-

A Process which takes place on its own without any outside assistance and moves towards an equilibrium state is called Spontaneous Process. OR A Reaction which needs energy to start but once it is started then it takes place on its own is called Spontaneous Process.

Example :-

Flow of water from high level to low level.

Flow of heat from hot body to cold body.

Spontaneous Process is Natural Unidirectional and irreversible.

NON Spontaneous Reaction :-

A Reaction which is Reverse of Spontaneous Reaction and does not take place on its own is called non spontaneous Reaction.

Example :-

Pumping of water uphill.

Freezing of water in Refrigerator.

Paper - 2021

(Semester 6th)

• SHORT Question • :-

Q.10

Write Mathematical Relation B/w entropy and Probability ?

Entropy :- Randomness or disorderness of a system is called entropy. it is denoted by "S"

Probability :- thermodynamic Probability is the number of microstate corresponding to a given macrostate when we dealing with distribution of molecule amongst an extremely large number of energy levels. it is denoted by "w"

Mathematical Relation :-

$$w \propto S$$

when Probability increases then entropy of system is also increases.

$$S = K \ln w$$

where K is Boltzman constant. this is the Relation B/w entropy and Probability.

Give At least Two Postulates of Bimolecular Collision Theory ?

- i) Reaction B/w two molecules takes place due to collisions B/w them.
- ii) All collision B/w the molecules do not lead to a chemical Reaction.
- iii) The collision which cause a chemical Reaction to take place are termed as effective or successful collision. The other being ineffective.
- iv) For the collision to be effective, the Reactant molecule must have energy greater than activation energy and proper Orientation so as to cause head on Collision.
- v) The Rate of Reaction is directly Proportional to the Number of Collision per unit volume per unit time, the Fraction of Molecule with Proper orientation and Fraction of Molecule having energy greater than Activation energy.

2iii)

Derive Average velocity ?

The Average velocity of molecule of gas is defined as the sum of all velocity of all molecules divided by the total number of molecules of gas.

$$\bar{c} = \frac{c_1 + c_2 + c_3 + \dots + c_n}{n} \Rightarrow \frac{1}{n} \sum c_i$$

$$\bar{c} = \frac{1}{n} \int_0^{\infty} c dn_c \Rightarrow \int_0^{\infty} \frac{c dn_c}{n}$$

Where dn_c is number of molecules having velocity "c"

We know that

$$\frac{dn_c}{n} = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-\frac{mc^2}{2kT}} c^2 dc$$

$$\bar{c} = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^{\infty} e^{-\frac{mc^2}{2kT}} c^3 dc$$

$$\bar{c} = 4\pi \left(\frac{1}{\alpha^2 \pi} \right)^{3/2} \int_0^{\infty} e^{-\frac{c^2}{\alpha^2}} c^3 dc \quad \left(\because \alpha^2 = \frac{2kT}{m} \right)$$

$$\bar{c} = 4\pi \left(\frac{1}{\alpha^2 \pi} \right)^{3/2} \cdot \frac{\alpha^4}{2} \quad \left(\because \int_0^{\infty} e^{-\frac{c^2}{\alpha^2}} c^3 dc = \frac{\alpha^4}{2} \right)$$

$$\bar{c} = \frac{4\pi}{\pi^{3/2}} \cdot \frac{\alpha^4}{2} \Rightarrow \frac{2\alpha}{\sqrt{\pi}}$$

$$\bar{c} = 2 \sqrt{\frac{2kT}{\pi m}} \Rightarrow \sqrt{\frac{8kT}{\pi m}}$$

$$\bar{c} = \sqrt{\frac{8RT}{\pi m}}$$

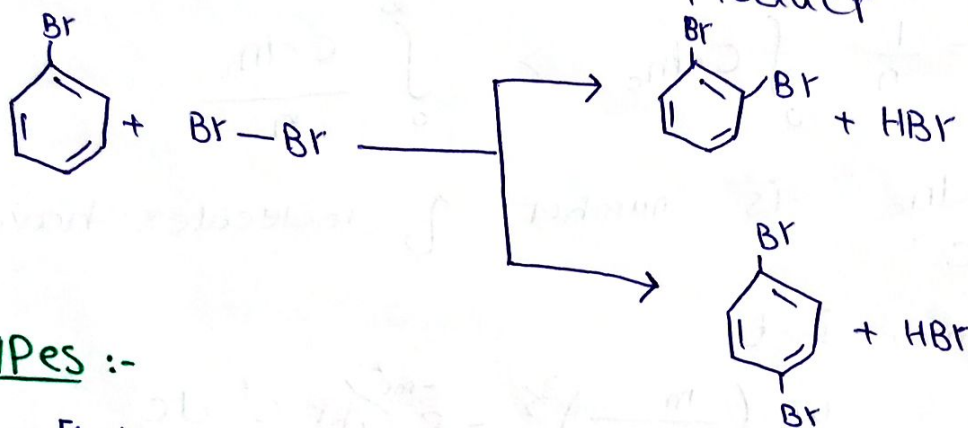
$$\bar{c} = 1.59 \sqrt{\frac{RT}{M}}$$

(iv)

What is Side Reaction give its types ?

In Parallel Reaction the Reactants gives two or more products by following two or more paths. Relatively high yield of product is obtained in one of the Reaction which is termed as main or major Reaction whereas the other Reaction are called Side Reaction. So that yield of main product is reduced.

Example :-



Types :-

- First order type of Parallel Reaction
- Second order type of Parallel Reaction

and

Describe Most Probable Velocity ?

The most Probable velocity is defined as the velocity possessed by the maximum number of molecules of gas at given Temperature. The expression for most Probable velocity can be derived mathematically by using the condition of Maxima by setting first derivative of $\frac{1}{n} \cdot \frac{dn}{dc}$ with respect to speed equal to zero, thus,

$$\frac{1}{n} \cdot \frac{dn}{dc} = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} \cdot \frac{-mc^2}{e^{2RT}} \cdot c^2 \rightarrow (A)$$

diff. eq (A) with respect to "c"

$$\frac{d\left(\frac{1}{n} \cdot \frac{dn_c}{dc}\right)}{dc} = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} \frac{d}{dc} \left[e^{-\frac{mc^2}{2RT}} c^2 \right]$$

$$= 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} \left[e^{-\frac{mc^2}{2RT}} \cdot 2c + c^2 \cdot e^{-\frac{mc^2}{2RT}} \cdot \frac{(-2Mc)}{2RT} \right]$$

$$= 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} \left[2c e^{-\frac{mc^2}{2RT}} - \frac{2Mc}{2RT} \cdot e^{-\frac{mc^2}{2RT}} \cdot c^2 \right]$$

$$= 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} \cdot 2c e^{-\frac{mc^2}{2RT}} \left[1 - \frac{Mc^2}{2RT} \right]$$

At Maxima Point

$$\frac{d\left(\frac{1}{n} \cdot \frac{dn_c}{dc}\right)}{dc} = 0$$

Hence Rearranging the terms

$$4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} 2c e^{-\frac{mc^2}{2RT}} \left[1 - \frac{Mc^2}{2RT} \right] = 0$$

there are three possibilities

i) $c = 0$
 ii) $e^{-\frac{mc^2}{2RT}} = 0$ when $c = \infty$

iii) $\left(1 - \frac{Mc^2}{2RT} \right) = 0$ when $c = \sqrt{\frac{2kT}{m}}$

The first two possibilities corresponding to minimum

when $c \rightarrow 0$, $dn_c \rightarrow 0$

when $c \rightarrow \infty$, $dn_c \rightarrow 0$

The third condition satisfy Maxima Point.

$$1 - \frac{Mc^2}{2RT} = 0$$

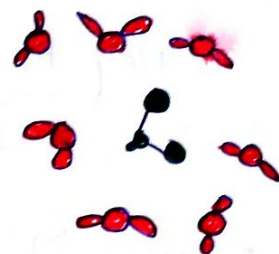
$$1 = \frac{Mc^2}{2RT} \Rightarrow c^2 = \frac{2RT}{M} \Rightarrow \boxed{C_{mp} = \sqrt{\frac{2RT}{M}} \Rightarrow 1.414 \sqrt{\frac{RT}{M}}}$$

Q. viii

What is cage effect ?

The cage effect in chemistry describes, How the properties of molecule are affected by its surroundings.

Capture the molecules of reactant diffuse in or out to other solvent cage,



Once reactant molecules meet in one solvent cage through diffusion they will be trapped in the cage for a relatively long time during which they collide repeatedly with each other providing more chance to react.

Within solvent cage (for 1-100 ns) two reactant molecules can collide $10 \sim 10^5$ times.

Cage effect is also the factor effect on rate of reaction, due to this reaction in solution phase are very fast than gas phase reaction.

Q. viii

Reaction in solution phase are faster than gas phase why ?

Reaction in solution phase are very fast than that of the gas phase due to solvent cage effect. Molecules of different reactant are in the cage of solvent and collide with each other many times and are converted into product in very short intervals of time but in gas phase reaction the probability of collision of molecules of reactant is less as compared to that in solution. This is the reason reaction in solution phase are faster than gas phase.